

NAZARBAYEV INTELLECTUAL SCHOOL OF PHYSICS AND MATHEMATICS IN AKTOBE

Бекітемін:
Мектеп директоры
Буксукбаев Қ.С.

« » _____ 2023 жыл

Келісілді:
Директордың оқу ісі жөніндегі
орынбасары
Кужакова С.Ж. _____
« » _____ 2023 жыл

Бірлестік мәжілісінде қаралды
Хаттама №01 « 29 » 08. 2023 жыл
бірлестік меңгерушісі
Мұқашева И.А. _____
« » _____ 2023 жыл

Yearly course calendar for chemistry 2023-2024 academic year

Subject: Chemistry
Grade: 12
Language of instruction: English
Hours: 192
Weeks: 32
Teacher:

№	Topic of the Lesson	Learning Objectives
1	2	3
1-Term	12.1A Group 14	
1.	Occurrence and extraction of the elements	12.2.1.1 know the occurrence of, and be able to compare and contrast, methods of extraction of the elements
2.	Physical trends within the group	12.2.1.2 recognise trends in chemical and physical properties down the group
3.	Chemical trends within the group	12.2.1.2 recognise trends in chemical and physical properties down the group
4.	Shapes of molecules of their compounds	12.2.1.3 be able to explain the shapes of the molecules of the compounds
5.	Trends in oxidation states within the group	12.2.1.4 know and understand trends in oxidation states down the group 12.2.1.5 know and be able to explain the properties of the oxides of oxidation states II and IV
6.	Laboratory work № 1 "Chemical properties of tin and lead and their compounds" Demonstration # 1 "Chemical properties of tin and lead oxides (SnO, SnO ₂ PbO, PbO ₂)". Chemical properties within a group	12.2.1.6 know and be able to explain the relative stability of the II and IV oxidation state in aqueous solution
	12.1B Nitrogen and sulfur	
1.	Nitrogen: • Reactivity Ammonia and its structure	12.2.1.7 understand the bonding and be able to explain the unreactivity of nitrogen 12.2.1.8 be able to draw and explain the shape of the ammonia molecule
2.	Laboratory work №2 "Obtaining ammonia and studying the properties of its aqueous solution"	12.2.1.9 know and understand the chemistry of ammonia as a gas and in aqueous solution
3.	Ammonia and its compound	12.2.1.10 know the industrial manufacture of ammonia and be able to understand it as an equilibrium process
4.	Environmental impact of oxides of nitrogen and nitrates Practical work №1 "The chemical properties of HNO ₃ » Demonstration №3 «The chemical properties of concentrated nitric acid"	12.2.1.11 know and understand the environmental impact of oxides of nitrogen in the atmosphere and nitrates in soils and water supplies
5.	Environmental impact of oxides of nitrogen and nitrates	12.2.1.11 know and understand the environmental impact of oxides of nitrogen in the atmosphere and nitrates in soils and water supplies

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	Practical work №2 "Determination of nitrate in drinking water and vegetables"	12.2.1.12 be able to evaluate ways of minimising these environmental impacts
6.	Allotropic modifications of sulfur Laboratory work №3 Physical properties of sulfur Demonstration # 4 "Sulfur burning in oxygen"	12.2.1.13 know the occurrence of sulfur 12.2.1.14 understand the structure and allotropy of sulfur
7.	Sulphur dioxide and acid rain	12.2.1.15 know the sources of sulfur dioxide in the atmosphere and understand its consequences, including the formation and effects of acid rain
8.	Sulphur dioxide and acid rain Practical work 3 "Chemical properties of H ₂ SO ₄ "	12.2.1.16 understand the need for food preservatives and know the use of sulfur dioxide for this purpose 12.2.1.17 understand the role of sulfuric acid as an acid and as an oxidising agent
9-10	Sulfuric acid. Contact process. Demonstration №5 «Chemical properties of conc. sulfuric acid "	12.2.1.18 recall at least one process for the manufacture of sulfuric acid in terms of an equilibrium process and know its major uses as an industrial chemical 12.2.1.19 understand the environmental impact of sulfuric acid manufacture
11-12.	Generalization and systematization of knowledge. Summative assessment for unit “12.1AGroup 14” and “12.1B Nitrogen and sulfur”	
	12.1C Acids and bases	
1-2.	Theory of acids and bases of Brønsted-Lowry Laboratory work №4 Properties of acids and bases	12.3.4.1 understand acidity/alkalinity in aqueous solution and that, in this medium, the approaches of Arrhenius and Brønsted-Lowry are equivalent 12.3.4.2 recognise that in aqueous solution acids are proton donors and bases are proton acceptors
3-4	pH: definition and determination Hydrogen index: definition and methods of its measurement	12.3.4.3 understand pH as $-\log_{10}[\text{H}^+]$, and be able to convert pH to concentration and concentration to pH 12.3.4.4 be able to calculate the pH of a strong acid
5-6	Ionic product of water	12.3.4.5 know that water is weakly dissociated 12.3.4.6 know the ionic product of water, K_w and that $K_w = [\text{H}^+][\text{OH}^-]$
7-8.	Acid and base strength	12.3.4.7 be able to calculate the pH of a strong base 12.3.4.8 understand that weak acids and bases dissociate slightly in water
9.	Laboratory work №5 Determination of pH for solutions of HCl, NaOH, CH ₃ COOH, NH ₄ OH with different concentrations	- be able to calculate pH for a strong base - Understand indicators as having a characteristic pH at their endpoint and be able to use this to choose an appropriate indicator for a titration - explain the choice of suitable indicators for acid-base titrations, given appropriate data
10-11.	K_a / K_b and pK_a / pK_b	12.3.4.9 be able to write and perform calculations with the acid dissociation constant K_a 12.3.4.10 know pK_a , pK_b and pK_w and be able to use the equation $pK_w = pK_a + pK_b$

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		(where K_b is the base dissociation constant)
12-13.	pH curves and indicators	12.3.4.11 understand how to obtain and work with pH curves for the various acid/base combinations 12.3.4.12 understand indicators as having a characteristic pH at their endpoint and be able to use this to choose an appropriate indicator for a titration
14.	Acid-base titrations Practical work №4 "Acid - base titration" of strong / weak acids and strong / weak bases	12.3.4.13 be able to carry out titrations and the associated calculations - be able to perform titration and make appropriate calculations -explain the choice of suitable indicators for acid-base titrations, given appropriate data
15.	Practical work №4 "Acid - base titration" of strong / weak acids and strong / weak bases	Be able to calculate the pH of a buffer solution Be able to calculate the pH of a buffer solution Describe the changes in pH during acid-base titrations and explain these changes in terms of the strengths of the acids and bases
16-17	Buffer solutions	12.3.4.14 understand the action of acidic and basic buffer solutions qualitatively in terms of equilibria 12.3.4.15 be able to calculate the pH of a buffer solution 12.3.4.16 know some uses of buffer solutions
18.	Practical work №5 "Preparation of buffer solutions and determination of their pH"	Be able to calculate the pH of a buffer solution Describe and explain the uses of buffer solutions, including the role of HCO_3^- in controlling pH in blood
19-20.	Generalization and systematization of knowledge. Summative assessment for unit "12.1C Acids and bases"	
	12.1D Nomenclature and isomerism of organic compounds.	
1-2	IUPAC nomenclature for aromatic and other complex compounds	12.4.2.1 understand and be able to work with IUPAC nomenclature extended to more complex compounds including cyclic, aromatic and heterocyclic compound
3-4.	Asymmetric carbon atoms, optical isomers and their effect on plane polarised light Demonstration № 6 "Construction of spherical models of optical isomers"	12.4.2.2 be able to draw and build structures which show geometrical (E-Z) and optical isomerism 12.4.2.3 know that optical isomers will rotate the plane of polarised light describe structural isomerism and its division into chain, positional and functional group isomerism

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5-6.	Enantiomers and racemates Demonstration # 7 "Influence of active molecules on the plane of polarized light and show the effects of a mixture with concentrations of both isomers (racemic mixture)	12.4.2.4 understand the effects of chemical reactions on stereoisomers, including racemisation
7-8.	Biological activity of optically active compounds	12.4.2.5 appreciate that stereoisomers may have different effects on living systems
9-10.	Generalization and systematization of knowledge. Summative assessment for unit "12.1D Nomenclature and isomerism of organic compounds"	
11-12.	Cambridge Revision Paper (Summative exam)	
13-14.	Cambridge Revision Paper (Summative exam)	
15-16.	Summative assessment for the term 1	
2-Term	12.2A Carbonyl compounds	
1	General characteristics of aldehydes and ketones	12.4.2.6 know and understand the structure of the carbonyl group 12.4.2.7 recognise and be able to name and write formulas and structures for, a range of carbonyl compounds
2.	Practical work № 6 "Obtaining aldehydes and ketones and studying their properties"	12.4.2.8 know the reaction of Tollen's reagent and Fehling's/ Benedict's reagent with aldehydes and ketones and understand their use in testing
3-4.	Oxidation, reduction and nucleophilic addition reactions	12.4.2.9 know and be able to account for the products of oxidation and reduction of aldehydes and ketones 12.4.2.10 understand and know examples of nucleophilic addition reactions of aldehydes and ketones
5.	General characteristics of carboxylic acids Acidity (Properties of carboxylic acids)	12.4.2.11 recognise carboxylic acids as weak acids and know their reaction with carbonates
6.	Laboratory work №6 «Preparation and properties of carboxylic acids	-the oxidation of methanoic acid, HCO_2H , with Fehling's and Tollens' reagents. -the oxidation of ethanedioic acid, $\text{HO}_2\text{CCO}_2\text{H}$, with warm acidified manganate(VII)
7.	The esterification reaction Ethers. General characteristics	12.4.2.12 know the esterification reaction and the hydrolysis reaction, recognise these as equilibria and determine appropriate conditions using equilibrium considerations
8.	Practical work №7 "Preparation of esters and their hydrolysis"	-formation of esters by esterification with carboxylic acids. -formation of esters by acylation with acyl chlorides using ethyl

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		ethanoate and phenyl benzoate as examples.
9.	Natural esters and soaps Practical work №8 "Getting soap"	12.4.2.13 know some uses of esters, including natural esters, soaps and biodiesel
10.	Acylation Demonstration № 8 "Properties of anhydrides, acetic acid chlorides"	12.4.2.14 recognise acid chlorides and acid anhydrides 12.4.2.15 understand their reactions and their importance in organic synthesis 12.4.2.16 know and be able to use the mechanism of acylation reactions 12.4.2.17 understand the role of acylation in the manufacture of aspirin
11.	Practical work №9 "Study of the properties of synthetic detergents"	
12.	Generalization and systematization of knowledge. Summative assessment for unit "12.2A Carbonyl compounds"	
	12.2B Transition metals	
1.	Definition of a transition metal General characteristics of transition metals	12.2.1.20 understand what is meant by a transition metal and be able to describe this in terms of electronic structure (Ti - Cu) 12.2.1.21 recognise the characteristics of transition metal chemistry
2	Laboratory work №7 "Properties of salts of transition metals" Demonstration №9 "Interaction of salts of transition metals with NH ₃ (conc)"	
3.	Ligands and complex formation	12.2.1.22 understand the general structure of a complex ion and the nature of ligands 12.2.1.23 know that ligands can be unidentate or multidentate and be able to identify examples of each
4.	Shapes of transition metal complexes	12.2.1.24 understand, be able to predict the shape of, and be able to draw octahedral, tetrahedral and linear complexes
5.	Formation of coloured ions Demonstration №10 "Determination of concentration of transition metal ions in solutions" Laboratory work №8 "Reaction of a traffic light"	12.2.1.25 understand the reasons for the colours and colour changes which complex ions undergo on reaction and be able to give a general description of this in terms of electronic transitions 12.2.1.26 understand the use of and be able to use spectrometry to determine concentration
6.	Laboratory work №9 "Equilibrium in solutions of salts of transition metals"	-predict from a given electronic configuration, the likely oxidation states of a transition element.
7.	Variability of oxidation states	12.2.1.27 know that transition elements show variable oxidation states and be able to describe specified examples

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		12.2.1.28 know that transition metals are generally good catalysts and be able to account for this in terms of variation of oxidation states
8.	Catalysis (homogeneous and heterogeneous) Demonstration № 11 "Autocatalysis. The reaction between potassium permanganate and oxalic acid » Laboratory work № 10 "Transition metals and their compounds - catalysts" Laboratory work No. 11 "Reaction of potassium peroxide and potassium iodide in the presence of iron (II) ions"	12.2.1.29 understand a general mechanism for heterogeneous catalysis and know specified examples 12.2.1.30 understand a general mechanism for homogeneous catalysis and know specified examples
9.	Biological role of transition metals	12.2.1.31 know that haemoglobin contains iron complexes and understand its role in oxygen transport, including carbon monoxide poisoning
1.	Generalization and systematization of knowledge. Summative assessment for unit “12.2B Transition metals”	
	12.2C Aromatic compounds	
1.	Bonding in benzene and its compounds	12.4.2.19 understand and be able to describe, using thermochemical evidence, the additional stability conferred by the structure
2.	Delocalization of 6 π -electrons	12.4.2.19 understand and be able to describe, using thermochemical evidence, the additional stability conferred by the structure
3.	Electrophilic substitution	12.4.2.20 understand and be able to explain why the structure causes benzene to undergo electrophilic reactions 12.4.2.21 know and understand a general mechanism for electrophilic substitution and be able to use it in discussing the chemistry of benzene
4-5.	Nitrates and nitrobenzenes	12.4.2.22 understand the mechanism of nitration and the importance of nitration in synthesis 12.4.2.23 know some compounds for which nitration is an important precursor
6.	Friedel-Crafts acylation	12.4.2.24 understand Friedel-Crafts acylation and its importance in synthesis
7.	Practical work №10 "Characteristic reactions of aromatic compounds"	-predict whether halogenation will occur in the side-chain or in the aromatic ring in arenes depending on reaction conditions. -recognise the potential for hydrogen bonding in proteins and understand its effects
8.	Generalization and systematization of knowledge. Cambridge Revision Paper	
	12.2D Amines and amino acids	
1.	Amines. Classification	12.5.1.1 know the classification of amines and understand that they act as Bronsted-

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		Lowry bases
2.	Basicity Basic nature	12.5.1.2 be able to explain the difference in base strength between ammonia, primary, secondary and tertiary amines
3.	Nucleophilic properties Demonstration No. 12 "Aniline Diazotization Reaction"	12.5.1.3 understand the mechanism of formation of amines from haloalkanes by nucleophilic substitution and from nitriles by reduction
4.	Laboratory work №12 "The reaction of amines and their comparison with the properties of ammonia"	- understand the mechanism of amine formation through nucleophilic substitution reactions of haloalkanes and by reduction of nitriles -describe the reduction of amides with LiAlH ₄ -describe the formation of amides from the reaction between NH ₃ or RNH ₂ and R'COCl -describe amide hydrolysis on treatment with aqueous alkali or acid
5.	Preparation of amines	12.5.1.4 know the preparation of aromatic amines by reduction of nitro compounds 12.5.1.5 know the uses of amines as synthetic intermediates
6	General characteristics of amino acids	12.5.1.6 understand what is meant by an amino acid and that they are capable of both acid and base behaviour 12.5.1.7 be able to use the systematic names of common amino acids and know their trivial names 12.5.1.8 understand that amino acids are usually optically active
7.	Acid / basic properties, bipolar ions	12.5.1.9 recognise the ability of amino acids to form zwitterions and understand the conditions in which this takes place
8.	Peptide bonding	12.5.1.10 understand the formation of peptide links and of proteins from amino acids
9.	Formation of proteins Laboratory work №13 "Separation of a mixture of amino acids by the method of paper chromatography"	12.5.1.11 recognise the potential for hydrogen bonding in proteins and understand its effects 12.5.1.12 know how to hydrolyse proteins and understand the products formed
10	Practical work №11 "Qualitative reaction to amino acids" "The reaction of ninhydrin" Van Slyck's reaction	
11.	Generalization and systematization of knowledge.	
12.	Cambridge Revision Paper "12.2C Aromatic compounds" and "12.2D	

Nº	Topic of the Lesson	Learning Objectives
	Amines and amino acids” Summative assessment for unit	
3-Term	12.3A Aqueous inorganic chemistry	
1-2.	Lewis acids and bases	12.3.4.17 recognise various acid-base theories (Arrhenius, Bronsted-Lowry and Lewis) and understand the conditions in which each may be used, their strengths and limitations
3.	Metal aqua-ions	12.3.4.18 recognise and know the structure of metal-aqua ions
4.	4.Practical work Nº12 "Chemical reactions of complex compounds of transition metals"	To understand the behavior and acid nature of these ions and their reactions with a solution of ammonia, solutions of hydroxides and solutions of carbonates to produce insoluble hydroxides and complex ions
5.	Acidity and hydrolysis	12.3.4.19 understand the acid behaviour of these ions and their reactions with ammonia solution, hydroxide and carbonate solutions to produce insoluble hydroxides and complex ions
6.	Mini project: "Hydrolysis of salts. Features of soil hydrolysis "," Hydrolysis of salts in the economy and everyday life of man "	
7-9.	Ligand substitution reactions	12.3.4.20 understand ligand substitution with similar and dissimilar sized ligands, and possible consequent changes of co-ordination number 12.3.4.21 be able to explain ligand substitutions in terms of complex ion stability 12.3.4.22 recognise the existence of bi- and multidentate ligands and understand the chelate effect in terms of entropy and free energy change
10.	Revision Unit 3A Cambridge Revision Paper	
	12.3B Synthetic polymers	
1.	.Addition polymerisation	12.4.2.25 understand that an organic compound with a C=C bond will polymerise by an addition mechanism 12.4.2.26 be able to draw the repeating unit of an addition polymer, a simplified block diagram of the chain and know an equation to represent the process 12.4.2.27 appreciate that the hydrocarbons used in addition polymerization are largely oil-based chemicals
2	Addition polymerisation	12.4.2.28 understand the mechanism of addition polymerisation 12.4.2.29 know a range of examples of addition polymers and some of their common uses
3-4.	Condensation polymerisation (polyamides and polyesters) 14.Condensation polymerisation (polyamides and polyesters)	12.4.2.30 know the potential for polymerisation of dicarboxylic acids and diols, dicarboxylic acids (and acid chlorides) and diamines 12.4.2.31 be able to write simple equations for representative polymerisation reactions
5-6.	Condensation polymerisation (polyamides and	12.4.2.32 know the linkages and repeating units for polyamides and polyesters

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	polyesters) Condensation polymerisation (polyamides and polyesters)	12.4.2.33 know a range of examples of condensation polymers and some of their common uses
7	Practical work №13 "Polymer Recognition"	- know the connections and structural links of polyamides and polyesters
8-9.	Environmental effects	12.4.2.34 understand the importance of synthetic polymers in modern life and know a range of common products which are made from them 12.4.2.35 understand why addition polymers are largely inert and understand the consequences of this for the environment 12.4.2.36 understand that condensation polymers can be hydrolysed and may thus be biodegradable
10	Impact on the environment Development of production of high-molecular compounds in Kazakhstan	12.4.2.37 know and understand how polymers may be disposed of 12.4.2.38 appreciate the advantages of recycling
11-12.	Revision Unit 3B Cambridge Revision Paper	
13-14	Generalization and systematization of knowledge. Summative assessment for unit "12.3AAqueous inorganic chemistry" and "12.3B Synthetic polymers"	
	12.3C Chemistry and materials design	
1-2.	Compounds with physiological activity	12.4.2.39 appreciate the range of compounds which have physiological actions 12.4.2.40 understand that many of these compounds occur in nature but can be synthesised 12.4.2.41 understand the importance of molecular shape and chirality in physiological action
3-4.	Design and synthesis of drugs	12.4.2.42 understand the importance of modern structure determination techniques in drug design 12.4.2.43 know some examples of synthetic drugs 12.4.2.44 understand the problems of drug delivery and some simple solutions
5-6.	Properties of polymers related to their structures Uses of polymers based on structure	12.4.2.45 know a range of properties of polymers and understand how these are related to their structure
7.	Nanotechnology: the nanoscale	12.4.2.46 appreciate the nanoscale
8-9.	Structure of carbon-based nanoparticles Laboratory work №14 "Estimation of nanoparticle sizes"	12.4.2.47 know and understand in outline the structure of some carbon-based nanoparticles (e.g. buckminsterfullerene, graphene nanotubes)
10.	Uses of nanoparticles	12.4.2.48 know some uses of nanoparticles

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11	Environment and energy How chemistry can provide solutions to some environmental problems	12.4.1.1 know and understand how chemistry can help to solve some environmental problems
12	Efficient use of natural resources	12.4.1.2 know and understand how chemistry can help to use natural resources more efficiently
13.	Revision Unit 3C Cambridge Revision Paper	
14.	Generalization and systematization of knowledge. Summative assessment for unit “12.3C Chemistry and materials design”	
	12.3D The chemistry of life	
1-2.	Factors governing the shape of proteins	12.5.1.13 know that proteins are polymers and be able to describe sequences in relation to properties 12.5.1.14 be able to determine the structure of a protein from hydrolysis data
3-4.	Structure of a polypeptide from data Laboratory work №16 "Hydrolysis of Protein"	12.5.1.15 distinguish between primary, secondary and tertiary protein structures and explain the occurrence of alpha helices and beta-pleated sheets 12.5.1.16 know the tertiary protein structure and recognise the forces which govern its shape
5.	Enzymes and their operation Laboratory work №17 "The action of the enzyme catalase and manganese oxide (IV) on the decomposition of hydrogen peroxide"	12.5.1.17 understand the process of enzyme catalysis and the specific action of enzymes in terms of a 'lock and key' model 12.5.1.18 recognise and understand competitive inhibition 12.5.1.19 understand denaturation and how it can be brought about
6	Practical work №14 "Color reactions of proteins"	
7.	Structure of DNA Laboratory work №18 "Manufacturing of RNA and DNA model"	12.5.1.20 know and understand a simple model of the structure of DNA including hydrogen bonding between base pairs 12.5.1.21 explain DNA encoding in outline for the amino acid sequence of proteins
8.	Practical work №15 "Isolation of Ribonucleoprotein from Yeast"	
9.	Mini project "Isolation of DNA from bananas and onions"	
10.	Genetic basis of disease	12.5.1.22 explain the chemistry of DNA mutation from provided data 12.5.1.23 discuss the genetic basis of disease in terms of altered base sequence, causing alterations in protein structure and function 12.5.1.24 explain how modification to protein/enzyme primary structure can result in new structure and/or function

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11-12	ATP and energy	12.5.1.25 describe using block diagrams the structure and hydrolysis of ATP and how this provides a source of energy
13-14.	Metals which are essential to life Heavy metal pollution in the environment Effects of heavy metals of proteins	12.5.1.26 know and describe simply the role of specified metals which are essential to life 12.5.1.27 know some sources of heavy metal pollution in the environment and how these metals enter the food chain 12.5.1.28 recognise that some metals are very toxic and be able to describe simply their effects on proteins - understand that some metals are very toxic, and be able to describe their effects on proteins
	12.3E Organic synthesis	
1.	Practical work №16 "Recognition of Organic Substances on the Basis of Qualitative Reactions"	- to understand the usefulness of some reactions for characterizing organic compounds or specific functional groups - be able to identify compounds containing specific functional groups and to identify compounds by chemical and physical tests.
2.	Identification of major functional groups	12.4.2.49 understand the usefulness of some reactions for the characterisation of organic compounds or specific functional groups 12.4.2.50 be able to identify compounds containing specific functional groups and identify compounds by means of chemical and physical tests
3.	Synthesising organic compounds by single reactions and sequences	12.4.2.51 understand the importance of synthesis in organic chemistry and recognise synthetically useful reactions 12.4.2.52 produce spider diagrams of synthetically useful reactions
4.	Practical work №17 "Have substance X in the chain of transformations and carry out reactions to carry out these transformations"	
5.	Calculation of yield and purity	12.4.2.53 solve problems in synthesis drawing on knowledge of the organic chemistry learned in the whole course
6.	.Practical work №18 "Obtaining bromoethane and calculating the yield of a product as a percentage of the theoretically possible"	Calculate the yield and purity of the product in organic reactions
7.	Practical work №19 "Electrolysis of solutions of substances»	
8.	Practical work №20 "Recognize the solutions of substances using qualitative reactions"	
9.	Revision Unit 3D,E	

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	Cambridge Revision Paper	
10.	Generalization and systematization of knowledge. Summative assessment for unit “ 12.3DThe chemistry of life ” and “ 12.4E Organic synthesis ”	
11-12	Cambridge Revision Paper (Summative exam)	
13-14.	Cambridge Revision Paper (Summative exam)	
	Term 4 (30 h)	
1-2	1.1A Atomic structure 1	
3-4	11.1B Amount of substance	
5-6	11.1C Bonding	
7-8	11.1D Periodicity	
9-10	11.2A Introduction to organic chemistry - alkanes	
11-12	11.2B Energetics	
13-14	11.2C Kinetics	
15-16	11.4D Thermodynamics	
17-18	11.2D Equilibria	
19-20	11.3A Alkenes and 11.3B Alcohols	
21-22	11.3C Haloalkanes and	
23-24	11.3D Analytical techniques	
25-26	11.4A Redox – Reactions	
27-28	11.4B Elements 17 groups (halogens)	
29-30	11.4C Group 2: The alkaline earth elements	